The Smart Home Glossary

Word	Explanation
3D Printer	A machine allowing the creation of a physical object from a three-dimensional digital model, typically by laying down many thin layers of a material in succession.
Automation	Home Automations allow you to automatically respond to things that happen in your Smart Home. You can turn the lights on at Sunset or pause the music when you receive a call, open the garage when you return back home, etc
AI	It stands for Artificial intelligence. It is a wide-ranging branch of computer science concerned with building smart machines capable of performing tasks that typically require human intelligence.
API	An IoT Application Programming Interface (API) is the tool software developers use to gather and transfer data from one application or computer to another. APIs enable developers to programmatically interact with software components both inside and outside of their own code.
Alexa	Alexa or Amazon Echo is a virtual assistant developed by Amazon. It is capable of voice interaction, music playback, making to-do lists, setting alarms, streaming podcasts, playing audiobooks, and providing weather, traffic, sports, and other real-time information, etc.
ARM	A type of architecture for computer processors, for example used on all the Raspberry Pi models.
Bandwidth	Bandwidth is defined as the maximum amount of data that can be transmitted over an internet connection in a specific range of time. In IoT bandwidth needs vary by use case.
Bitcoin	Bitcoin uses peer-to-peer technology to operate with no central authority or banks; managing transactions and the issuing of bitcoins is carried out collectively by the network.
Cast	Play video, music, or other content on another display in the same room.
CLI	Command Line Interface. A screen or terminal where we can only use Linux commands to interact with the operating system.
CPU	Central Processing Unit. The processor, the primary component of a computer to run everything.
DHCP	Dynamic Host Configuration Protocol. Networking service which automatically assign an IP address to each new device on the network.
Discord	Discord is a messenger allowing you to talk over voice, video, and text.
DNS	Domain Name System. A system or service which translates domain names to IP addresses.

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Word	Explanation
Docker	Docker is an open platform for developing, shipping, and running applications. Docker enables you to separate your applications from your infrastructure so you can deliver or run software quickly.
ESP32	ESP32 is a series of low-cost, low-power system on a chip microcontrollers with integrated Wi-Fi and dual-mode Bluetooth.
ESP8266	The ESP8266 is a low-cost Wi-Fi microchip, with built-in TCP/IP networking software, and microcontroller capability. It is the predecessor of ESP32 but it is slower and without Bluetooth.
ESPHome	ESPHome is a tool which aims to make managing your ESP boards as simple as possible. It reads in a YAML configuration file (just like Home Assistant) and creates custom firmware which it installs on your ESP device.
Firmware	The basic software controlling the low-level operations for a specific hardware.
Flash	The action to copy an operating system/firmware/custom code/ to a SD card or a IoT device like ESP32 or ESP8266 (ex. D1 Mini, Sonoff Basic)
GPIO	General Purpose Input Output. Can be found for example on the Raspberry Pi which include 40 GPIO pins. It is mainly used to create an electronic circuit and use extensions like sensors, cards, hardware attached on top (HAT), etc.
GUI	Graphical User Interface. Kind of opposite of CLI. Mouse and graphical tools are available to make the device management easier.
HACS	Home Assistant Community Store HACS gives you a powerful UI to handle downloads of all your custom needs.
HAT	Hardware Attached on Top. For example: extension cards that can be plugged on the GPIO ports of a Raspberry Pi.
HDMI	High-Definition Multimedia Interface. There are Mini or Micro-HDMI variants as well.
Headless	A term used to define the use of a Raspberry Pi without any screen.
Hostname	A name assigned to a device on a network.
Home Assistant	Home Assistant is a Python-based open-source smart home system. It supports smart home devices of multiple brands, ensuring easy voice control and automation of devices.
Internet Security Protocol (IPSec)	IPSec is a set of protocols between two points on the IP network that provide data authentication, integrity, encryption, decryption, and confidentiality.

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Word	Explanation
IoT	Internet of Things. IoT devices are pieces of hardware, such as sensors, relays, gadgets, appliances, or machines, that are programmed for certain applications and can transmit data over the internet or other networks.
IoT cloud	IoT cloud platforms are end-to-end services that provide the infrastructure for transmitting, computing, and storing data from IoT devices.
IoT connectivity	IoT connectivity links all of the elements of an IoT ecosystem and allows devices to send and transmit data. Common IoT connectivity types include cellular, Wi-Fi, Ethernet, Bluetooth, RFID, Z-Wave, ZigBee, Thread and more.
IP Address	Unique identifier for a device on a network. For example: 192.168.0.1
IR	Infra Red.
LAN	Local Area Network. Generally refers to your network at home. Opposite to a WAN (Wide Area Network) that we use to speak about the Internet.
Latency	Latency measures how long it takes for data to travel from one point to another. Low latency means data is transferred quickly between device, network, and the intended outcome.
Linux	A family of open-source operating system using the Linux kernel, the base of all the Linux distributions.
MAC Address	Media Access Control Address. A unique identifier assigned to each network card. Can be used in a DHCP server to reserve an IP address to each device. For example: 00:01:02:03:04:05
Matter	Matter is a not yet released open standard with a promise of reliable, secure connectivity where connected devices will work seamlessly together. It is a collaboration coming from Apple, Google, Amazon, Zigbee and many more.
MQTT	MQTT stands for Message Queueing Telemetry Transport and it is a connectivity protocol on top of TCP/IP designed for Internet of Things (IoT) devices. It allows extremely lightweight and simple, yet effective publish/subscribe communication.
Nabu Casa	Nabu Casa is a paid subscription service which is linked to your Home Assistant instance, the main purpose of Nabu Casa is two fold, give the smart home owner access to their smart home outside of their home network, and secondly to help fund the home assistant developers and cloud infrastructure.

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Word	Explanation
Node-RED	Node-RED is a programming tool for wiring together hardware devices, APIs and online services. It provides a browser-based editor that makes it easy to wire together flows using the wide range of nodes in the palette that can be deployed to its runtime in a single-click.
NVR	A network video recorder is a specialized computer system that includes a software program that records video in a digital format to a disk drive, USB flash drive, SD memory card or other mass storage device.
ONVIF	ONVIF is an open industry forum that provides and promotes standardized interfaces for effective interoperability of IP-based physical security products like security cameras and NVRs.
OS	A software that manages everything on a computer (hardware, resources, software, services). For example: Windows, macOS, Linux, iOS, Android.
PSU	Stands for Power Supply Unit.
RAM	Random Access Memory. A temporary and fast storage type present on any computer. In general, the more RAM you have, the faster your programs will run. It's also better to use several apps simultaneously.
Raspberry Pi	The Raspberry Pi is a low cost, credit-card sized computer that plugs into a computer monitor or TV, and uses a standard keyboard and mouse. There is also a model (Raspberry Pi 400) with embedded keyboard.
Raspberry Pi OS	A Linux distribution especially tailored for the Raspberry Pi. It's the default operating system, based on Debian.
Repository	A server or group of servers on the Internet hosting software files used by package managers or Git clients.
RF	Radio Frequency.
root	The name of the administrator account on Linux systems.
RTSP	The Real Time Streaming Protocol is a network control protocol designed for use in security, entertainment and communications systems to control streaming media. The protocol is used for establishing and controlling media sessions between endpoints. For example between Security Camera and computer.
Samba	Samba is the standard Windows interoperability suite of programs for Linux and Unix. Samba provides file and print services for various of clients.
SD Card	Secure Digital card (microSD card in fact). This is the default and main storage device on Raspberry Pi. Can be used in phones and smart security cameras for example.

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Word	Explanation
SSD	Solid-State Drive. A storage device, faster than the usual HDD, and also the SD cards. Can be used as the main storage on recent Raspberry Pi models (instead of the SD card).
SSH	Secure Shell Protocol. A network protocol used to remotely access a computer.
sudo	It stand for "superuser do", as originally that is all it did. It is actually a program for Unix-like computer operating systems that enables users to run programs with the security privileges of another user, by default the superuser or root.
Tasmota	Tasmota is an open source firmware for Espressif ESP8255, ESP32, ESP32-S or ESP32-C3 chipset based devices.
Telegram	Telegram is a messaging app with a focus on speed and security, it's super-fast, simple and free.
TTS	Text-to-speech (TTS) is a type of assistive technology that reads digital text aloud.
VM	A virtual machine (VM) is a virtual environment that functions as a virtual computer system with its own CPU, memory, network interface,
VPN	A virtual private network (VPN) gives you online privacy and anonymity by creating a private network from a public internet connection.
Wi-Fi	Wi-Fi is the wireless technology used to connect computers, tablets, smartphones and other devices only locally or to the internet or both.
WoL	Wake-on-LAN (WoL) is an Ethernet or Token Ring computer networking standard that allows a computer to be turned on or awakened by a network message.
ZigBee	Zigbee is a low-cost, low-power, wireless mesh network standard targeted at IoT, battery-powered devices in wireless control and monitoring applications.
Z-Wave	Z-Wave is a wireless communications protocol used primarily for home automation. It is a mesh network using low-energy radio waves to communicate from appliance to appliance,[1] allowing for wireless control of residential appliances and other devices, such as lighting control, security systems, thermostats, windows, locks, swimming pools, and garage door openers.

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Word	Explanation
Home Assistant Related Terms (Info from the Official HA Docs)	
Action	Actions are used in several places in Home Assistant. As part of a script or automation, actions define what is going to happen once a trigger is activated. In scripts, an action is called sequence. Actions use service calls and/or scenes to interact with entities and cause these entities to do something. Actions can also include conditions and a delay. An action can call multiple services at the same time. For example, if your presence is detected in a room, an action may call one service to turn on a light and call another service to start playing music after a delay. Actions are also used on the dashboard, for example as tap or hold action on a UI element. When triggered, the action calls a service.
Add-on	Add-ons are additional standalone third-party software packages that can be installed on Home Assistant OS. Most of these, add-on provided, applications can be integrated into Home Assistant using integrations. Examples of add-ons are: an MQTT broker, database service or a file server.
Area	An area in Home Assistant is a logical grouping of devices and entities that are meant to match areas (or rooms) in the physical world: your home. For example, the living room area groups devices and entities in your living room. Areas allow you to target service calls at an entire group of devices. For example, turning off all the lights in the living room. Locations within your home such as living room, dance floor, etc. Areas can be assigned to floors. Areas can also be used for automatically generated cards, such as the Area card.
Automation	Automations connect one or more triggers to one or more actions in a 'when trigger then do action' fashion with additional optional conditions. For example, an automation might connect the trigger 'sunset' to the action 'turn the lights on' but only if the condition 'someone is home' is met. Pre-made automations for common use-cases are available via the blueprints feature.
Backup	Home Assistant has built-in functionality to create files containing a copy of your configuration. This can be used to restore your Home Assistant as well as migrate to a new system. The backup feature is available on some installation types.
Binary sensor	A binary sensor returns information about things that only have two states - such as on or off.

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Word	Explanation
Blueprint	A blueprint is a script or automation configuration with certain parts marked as configurable. This allows users to create multiple scripts or automations based on the same blueprint, with each having its own configuration-specific settings. Blueprints are shared by the community on the blueprints exchange in the forum.
Category	A category is an organization tool that allows grouping items in a table. Like labels, categories allow grouping irrespective of the items' physical location. For example, on the automations page, you can create the categories "Notifications" or "NFC tags" to view your automations grouped or filtered. Categories are unique for each table. The automations page can have different categories than the scene, scripts, or helpers settings page.
Commissioning	In the context of Matter devices, commissioning is the process of adding a device to a Matter controller. It is the equivalent of pairing a device in Zigbee or Z-Wave. Commissioning is done by scanning a QR code or entering a code manually. The code is printed on the device or its packaging. The code contains information about the device, such as its type, manufacturer, and serial number. The controller uses this information to identify the device and to download the required information to control the device. For example, the controller downloads the device's capabilities, such as the supported commands and the available attributes. The controller also downloads the device's configuration, such as the device's name and location.
Component	Better known as: Integrations. Integrations used to be known as components.
Condition	Conditions are an optional part of an automation that will prevent an action from firing if they are not met.
Configuration file	The configuration.yaml file is the main configuration file for Home Assistant. It lists the integrations to be loaded and their specific configurations. In some cases, the configuration needs to be edited manually directly in the configuration.yaml file. Most integrations can be configured in the UI.
Cover	Covers are devices such as blinds, garage doors, etc that can be opened and closed and optionally set to a specific position.
Customize	Customization allows you to overwrite the default parameters of your devices in the configuration.

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Word	Explanation
Device	A device is a model representing a physical or logical unit that contains entities. Example for a device as a physical unit A smart plug named ‘Coffee machine’ which provides 2 entities: a switch entity to turn power on or off (‘Coffee machine power switch’) and a sensor entity for power monitoring (‘Coffee machine power sensor’). Example for a device as a logical unit An ecobee thermostat with 4 room sensors. This thermostat is seen as 5 devices in Home Assistant: 1 device for the thermostat with 4 sensors, and 1 device for each room sensor. Each device can be in a different area and may have more than one input or output within that area. Devices have properties such as ID, manufacturer, name, model, hardware version, firmware version, connections, etc.
Device tracker	Device trackers are used to track the presence, or location, of a device.
Discovery	Discovery is the automatic setup of zeroconf/mDNS and uPnP devices after they are discovered.
Domain	Each integration in Home Assistant has a unique identifier: a domain. All of the entities and services available in Home Assistant are provided by integrations and thus belong to such a domain. The first part of the entity or service, before the . shows the domain they belong to. For example light.kitchen is an entity in the light domain from the light integration, while hue.activate_scene is the activate_scene service for the hue domain which belongs to the Hue integration.

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Word	Explanation
Entity	An entity represents a sensor, actor, or function in Home Assistant. Entities are used to monitor physical properties or to control other entities. An entity is usually part of a device or a service. Entities have states. Example for entities as part of a device A combined temperature and humidity sensor device provides two sensor entities. One for temperature (e.g. sensor.temperature with state 21.0 and unit °C) and one for humidity (e.g. sensor.humidity with state 65.4 and unit %). Example for entities as part of a service A weather service that provides 3 entities: wind speed, air pressure, and ozon level. Example of an entity used for control A fan that is turned on when the temperature exceeds 30 °C. There are standardized types of entities for common integrations such as light, switch, camera, sensor, fan, or vacuum. Some entities are not part of a device or service. Examples of standalone entities are automation, script, scene entities, and helper entities (e.g. input helpers). Most properties of entities are related to the state. Entities have optional attributes such as friendly name, unit of measurement, and an icon or picture that can be displayed in the frontend.
Event	An event is when something happens.
Floor	A floor in Home Assistant is a logical grouping of areas that are meant to match the physical floors in your home. Devices & entities are not assigned to floors but to areas. A floor has properties such as: Floor ID, name, aliases (for use in assistants), an icon, and a floor level. Some of these properties are optional. The level number can be negative to reflect floors below the basement. Floors can be used in automations and scripts as a target for actions. For example, to turn off all the lights on the downstairs floor when you go to bed.
Frontend	The frontend is a necessary component for the UI, it is also where you can define your themes.
Group	Groups are a way to organize your entities into a single unit.
HASS	HASS or hass is often used as an abbreviation for Home Assistant. It is also the command-line tool.
HassOS	Another name for Home Assistant Operating System
Home Assistant Container	Home Assistant Container is a standalone container-based installation of Home Assistant Core. Any OCI compatible runtime can be used, but the documentation focus is on Docker.

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Word	Explanation
Home Assistant Core	Home Assistant Core is the Python program at the heart of Home Assistant. It is part of all of the four installation types and can be run on various operating systems. It can be installed standalone (without Home Assistant Supervisor), using a Virtual Environment (typically referred to as Home Assistant Core installation method) or as a container using Docker (typically referred to as Home Assistant Container installation method).
Home Assistant Operating System	Home Assistant OS, the Home Assistant Operating System, is an embedded, minimalistic, operating system designed to run the Home Assistant ecosystem on single board computers (like the Raspberry Pi) or Virtual Machines. It includes Home Assistant Core, the Home Assistant Supervisor, and supports add-ons. Home Assistant Supervisor keeps it up to date, removing the need for you to manage an operating system. Home Assistant Operating System is the recommended installation method for most users.
Home Assistant Supervised	Home Assistant Supervised is a full UI managed home automation ecosystem that runs the Home Assistant Core program, the Home Assistant Supervisor and add-ons. It comes pre-installed on Home Assistant OS, but can be installed standalone on Debian based Linux distributions. It is one of the four supported installation methods, and for advanced users only.
Home Assistant Supervisor	The Home Assistant Supervisor is part of the Home Assistant OS and Home Assistant Supervised installation types and is responsible for system management. It is what makes an installation of Home Assistant a 'full' installation and is the piece of Home Assistant that controls the Home Assistant ecosystem. It is responsible for managing your system and is required for features including add-ons, snapshots, and updates.
Home Assistant Yellow	Home Assistant Yellow is the easiest way to run Home Assistant. Home Assistant Yellow comes in three variants: Home Assistant Yellow Kit (no Compute Module 4 included), Home Assistant Yellow (with an onboard wireless module), and Home Assistant Yellow PoE (with an onboard wireless module and Power over Ethernet support). They can also be purchased with or without the CM4. Home Assistant Yellow is designed to be user-friendly and easy to set up, making it a great choice for new users. It is also designed to be expandable, allowing users to add new features and functionality as their needs change.
Hub	A hub connects devices and/or services.
ID	IDs in Home Assistant are case sensitive.

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Word	Explanation
Integration	An integration is what connects Home Assistant to third-party systems and services. They are sometimes referred to as components. Integrations use services and/or the device and entity model to expose functionality in Home Assistant. Integrations have properties such as: Integration ID, name, and optional options. Examples of integrations are Zigbee Home Automation (ZHA), Hue, Deconz, ESPHome, and Ecobee. Some integrations can connect to multiple devices or services, for example, the MQTT integration can connect to multiple MQTT brokers.
Job	Jobs are the way to track scheduled work within Home Assistant. Jobs are usually created as part of a service call and are handled by the job scheduler. Examples of jobs are: turning on a light, sending a notification, or calling an external service. Jobs have properties such as ID, name, and status.
Label	Labels are an organization tool that allows grouping items in a table irrespective of the items' physical location. For example, on the automations page, you can create the labels "Morning routine" or "Lights" to view your automations grouped or filtered.
Local media	Local media is media that is stored on the device running Home Assistant, for example music or videos. Local media can be played on media players that are integrated into Home Assistant. The Local Media Browser allows you to manage media on your Home Assistant instance and in some cases even copy media from other sources to your local media storage.
Matter	Matter is a new standard for smart home devices developed by the Connectivity Standards Alliance (CSA). Matter devices are designed to work with any Matter controller, regardless of the manufacturer. This means that you can mix and match Matter devices from different brands and still have them work together. Matter devices are also designed to be easy to set up and use. For example, you can add a Matter device to your home network by scanning a QR code or entering a code manually.
Media browser	The media browser allows you to browse and play media from supported media sources, such as local media, online streaming services, or media players integrated into Home Assistant. The media browser is accessible from the sidebar in the Home Assistant frontend.

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Word	Explanation
Option	Options are a way to configure an integration. Options are specific to each integration and are usually set in the integration's options flow or YAML configuration. Examples of options are: polling interval, scan interval, and default volume. Options can also be used to configure entities or services. For example, you can set the default volume for a media player or the default temperature for a thermostat.
Passive	Passive devices are devices that do not actively send updates to Home Assistant. Instead, Home Assistant has to poll these devices to get their current state. Examples of passive devices are some types of temperature sensors or battery-powered devices that do not have a constant connection to the network. Passive devices are typically more battery efficient than active devices.
Person	A person in Home Assistant is a model representing a human that interacts with the system. Persons have properties such as: ID, name, and optional options. Examples of properties are: ID, name, user ID (for linking the person to a Home Assistant user), and device trackers (for tracking the person's location). Persons can be used in automations and scripts as a target for actions. For example, to send a notification to a person when they arrive home.
Scene	A scene is a predefined set of states for a group of entities. For example, you can create a scene called 'Movie night' that turns on the TV, dims the lights, and sets the temperature to 22 °C. Scenes can be activated manually or as part of an automation.
Script	A script is a sequence of actions that can be run manually or as part of an automation. Scripts are similar to automations but do not have triggers or conditions. Scripts can be used to group multiple actions together and to reuse them in multiple places. For example, you can create a script called 'Goodnight' that turns off all the lights and locks the doors. You can then call this script from an automation or from the frontend.
Service	Services are the way to control entities and devices in Home Assistant. Services are provided by integrations and can be called manually or as part of an automation or script. Examples of services are: light.turn_on, media_player.play_media, and notify.send_message. Services can have parameters, such as the entity ID to target or the message to send.
Snapshot	See backup.

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Word	Explanation
State	The state is the current value or condition of an entity in Home Assistant. States can be used in automations and scripts to trigger actions or to conditionally execute actions. For example, you can create an automation that turns on the lights when the state of a motion sensor changes to 'detected'.
Supervisor	See Home Assistant Supervisor.
Template	Templates are used in Home Assistant to dynamically generate text or perform calculations. Templates are written in the Jinja2 template language and can be used in many places, such as in automations, scripts, and the frontend. For example, you can use a template to send a notification with the current temperature or to calculate the average power consumption of your devices.
Trigger	Triggers are the way to start automations in Home Assistant. Triggers can be based on events, states, or time. Examples of triggers are: when a motion sensor detects movement, when the sun sets, or when a specific time is reached. Triggers can have conditions, which must be met for the trigger to activate the automation. For example, you can create a trigger that only activates when the sun sets and someone is home.
User	A user in Home Assistant is a model representing a person that can log in to the system. Users have properties such as: ID, name, username, and password. Users can have different roles, such as administrator or user, which determine their permissions. Users can be linked to persons to track their location and to send notifications.
View	A view is a part of the Home Assistant dashboard that displays a set of cards. Views can be used to organize your dashboard into different sections. For example, you can create a view for your living room, a view for your garden, and a view for your automations. Views can have different layouts, such as grid or masonry, and can be customized with themes and background images.
Webhook	Webhooks are a way for third-party services to send data to Home Assistant. Webhooks can be used to trigger automations or to update entities. For example, you can use a webhook to update a sensor with data from an online service or to trigger an automation when a specific event occurs. Webhooks can be configured in the Home Assistant frontend or via YAML configuration.

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Word	Explanation
Zone	A zone in Home Assistant is a geographical area that can be used to track the location of devices and persons. Zones can be used in automations and scripts to trigger actions or to conditionally execute actions. For example, you can create a zone for your home and an automation that sends a notification when you enter or leave the zone. Zones can have properties such as: ID, name, latitude, longitude, radius, and icon.